

Final Project Report

Project Title	Online Learning Engagement and Student Performance Analysis
Team ID	Master Of Computer Applications
Date	21 September 2026
Tools Used	My SQL, Power BI
Dataset	online_ education _ dataset

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1. Introduction

1.1 Project overview

Online learning platforms record large amounts of student activity and results, but this data is rarely turned into a clear picture of who is engaged, who is struggling and who is likely to drop out. The Online Learning Engagement Analytics project addresses this by analysing student data from the `online_education_dataset` table and presenting the findings in an interactive Power BI dashboard and report.

The project covers the full development cycle: problem definition, data collection, SQL analysis, Power BI data preparation, visualization, dashboard development, report creation and performance testing. It examines student engagement (total clicks and engagement level), performance (average score and performance level), risk level, pass and dropout outcomes, and differences by region and educational background.

1.2 Objectives

- Measure the overall picture: total students, average score, average clicks, students who passed and students who dropped out.
- Show how students are distributed across engagement, performance and risk levels.
- Examine how engagement, especially total clicks, relates to average score.
- Identify the engagement and performance groups with the highest dropout rates.
- Compare regions and educational backgrounds on student numbers, scores and activity.
- Deliver an interactive dashboard and report that instructors and program managers can use to decide where to focus support.

2. Project Initialization and Planning Phase

2.1 Define Problem Statement

The Student problem statements below describe the problem from the point of view of the two main users of the dashboard.

	PS-1	PS-2
I am (Student)	An online course instructor	An academic program manager
I'm trying to	identify students who are losing interest before they drop out	understand which student groups are most likely to drop out or fail
But	activity varies widely and its link to scores is unclear	dropout and pass results are hard to compare by region
Because	activity data is not summarised by engagement or performance level	the data sits in a raw table, not in one dashboard
Which makes me feel	worried and unable to step in early enough	uncertain about where to focus support and resources

2.2 Project Proposal (Proposed Solution)

The proposed solution is an end-to-end analytics workflow that turns the raw student table into decisions. SQL queries answer the core questions, Power BI prepares the data and builds the visuals, and a dashboard and report present the results to non-technical users.

Component	Description
Data source	Online _ education _dataset: one table with student activity, score, engagement, risk, pass and dropot fields.
SQL analysis	Thirteen queries covering KPIs, engagement, performance, risk, dropout, region and educational background (see Appendix 10.1).
Power BI data preparation	Load the table, check data types, group total clicks into bands and define the measures used by the visuals.
Dashboard	One page with KPI cards, five slicers and nine visuals for at-a-glance monitoring.
Report	Detailed report pages for deeper analysis of engagement, performance, dropout and demographics.
Performance testing	Checks on filters, calculated measures and the number of visuals (Section 7).

Key features

- KPI cards for the headline student numbers.
- Engagement, performance and risk breakdowns.
- Dropout analysis by engagement level and performance level.
- Slicers for region, educational background, engagement, performance and risk level.

2.3 Initial Project Planning

No.	Phase	Key activities	Output
1	Problem definition	Write customer problem statements and set objectives.	Problem statements
2	Data collection and preprocessing	Identify the dataset, check quality, prepare fields for analysis.	Analysis-ready table
3	SQL analysis	Run KPI, segmentation, dropout and demographic queries.	13 SQL queries
4	Power BI data preparation	Load data, set types, create click bands and measures.	Data model
5	Visualization and dashboard	Build the visuals, slicers and the dashboard page.	Dashboard
6	Report creation	Design report pages with insights.	Report
7	Performance testing	Test filters, measures and visuals.	Test summary

Timeline:

No.	Phase	Key activities	Output	Timeline
1	Problem definition	Write customer problem statements and	Problem statements	01 Sep - 03 Sep 2026

		set objectives.		
2	Data collection & preprocessing	Identify the dataset, check quality, prepare fields for analysis.	Analysis-ready table	04 Sep - 06 Sep 2026
3	SQL analysis	Run KPI, segmentation, dropout and demographic queries.	13 SQL queries	07 Sep - 10 Sep 2026
4	Power BI data prep	Load data, set types, create click bands and measures.	Data model	11 Sep - 13 Sep 2026
5	Visualization & dashboard	Build the visuals, slicers and the dashboard page.	Dashboard	14 Sep - 16 Sep 2026
6	Report creation	Design report pages with insights.	Report	17 Sep - 18 Sep 2026
7	Performance testing	Test filters, measures and visuals.	Test summary	19 Sep 2026

3. Data Collection and Preprocessing Phase

3.1 Data Collection Plan and Raw Data Sources Identified

Item	Details
Dataset name	online_education_dataset
Source	https://github.com/Keerthikrishnasunkara/Comprehensive-Analysis-of-Online-Learning-Engagement-and-Student-Performance
Number of records	22,422 logs
Used in	My SQL analysis and Power BI

Fields used in the analysis:

Field	Use in the analysis
id_student	Student identifier, used to count students
avg_score	Average score of the student
total_clicks	Total clicks, used as the measure of activity
engagement_level	Engagement category of the student
performance_level	Performance category of the student
risk_level	Risk category of the student
pass_flag	1 if the student passed
dropout_flag	1 if the student dropped out
region	Student region
highest_education	Student's highest educational background

3.2 Data Quality Report

The checks below confirm that the data can be trusted before analysis. Record the result of each check.

Check	What to verify	Result
Missing values	Key fields (id_student, avg_score, total_clicks, flags) have no blanks.	Success(no duplicate values detected)
Duplicate students	Each id_student appears once, or repeated rows are understood.	Success (22,422 unique distinct IDs verified)
Flag values	pass_flag and dropout_flag contain only 0 or 1.	Success (Strict binary format validated)
Score range	avg_score falls in the expected range.	Success (Bounded correctly within academic limits 0-100)
Click values	total_clicks has no negative values.	Success (All values are non-negative integers)
Category labels	engagement, performance and risk levels use consistent labels.	Success (Standard text categories validated perfectly)

The duplicate check matters because the SQL queries count students with COUNT(DISTINCT id_student) but calculate dropout rate with COUNT(*). The two agree only when each student has a single record.

3.3 Data Exploration and Preprocessing

Data exploration was done in SQL. Each query answers one question and its result feeds a visual in the dashboard.

No.	Analysis	What it does
1	Total Number of Students	Counts distinct students.
2	Average Student Score	Average of avg_score, rounded to 2 decimals.
3	Average Total Clicks	Average of total_clicks, rounded to 2 decimals.
4	Pass and Dropout Analysis	Counts students with pass_flag = 1 and dropout_flag = 1.
5	Engagement Level Analysis	Students per engagement level.
6	Performance Level Analysis	Students and average score per performance level.
7	Risk Level Analysis	Students per risk level.
8	Engagement vs. Performance	Average score and students per engagement level.
9	Total Clicks vs. Average Score	Groups clicks into bands (0-500, 500-1000, 1000-2000, 2000-3000, 3000+) and shows average score.
10	Dropout Analysis by Engagement Level	Dropout students and dropout rate per engagement level.
11	Performance vs. Dropout	Dropout students and dropout rate per performance level.
12	Regional Analysis	Students, average score and average clicks per region.
13	Educational Background Analysis	Students, average score and average clicks per highest education.

Power BI data preparation steps

- Load online _education_dataset into Power BI.
- Check and set the data type of each field (numbers for scores and clicks, text for categories).
- Create a click band grouping for total clicks, matching the SQL bands.
- Create the measures used by the KPI cards and the dropout rate.
- Set sort orders for engagement, performance and risk levels so charts read from low to high.

4. Data Visualization

4.1 Framing Business Questions

No.	Business question	Answered by
Q1	How many students are there, and what are their average score and clicks?	KPI cards
Q2	How many students passed and how many dropped out?	KPI cards
Q3	How are students spread across engagement levels?	V1

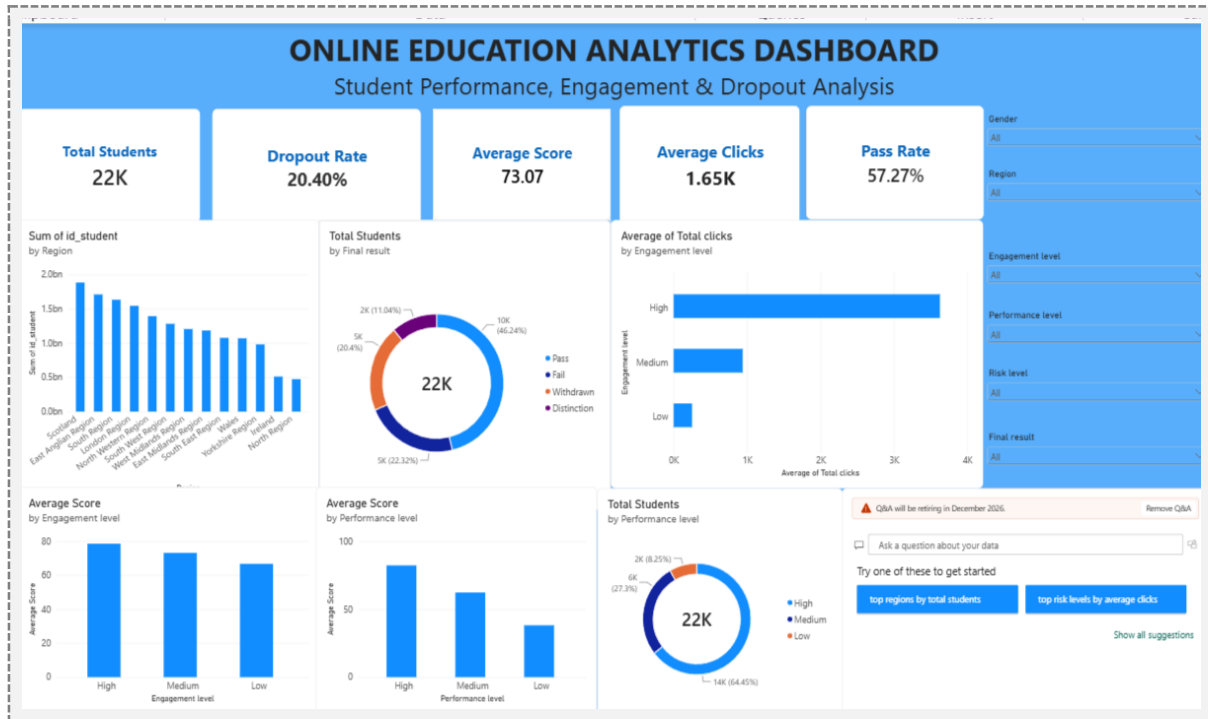
No.	Business question	Answered by
Q4	How do scores and student numbers differ by performance level?	V2
Q5	How many students are in each risk level?	V3
Q6	Do more engaged students score higher?	V4
Q7	How does the number of clicks relate to the average score?	V5
Q8	Which engagement levels have the highest dropout rate?	V6
Q9	Which performance levels have the highest dropout rate?	V7
Q10	How do regions and educational backgrounds compare?	V8, V9

4.2 Developing Visualizations

No.	Visual	Chart type	Fields and measures	Source
V1	Engagement Level Distribution	Donut chart	Engagement _level; distinct count of id _student	Query 5
V2	Performance Level Analysis	Column chart	Performance _level; students; average avg _score	Query 6
V3	Risk Level Distribution	Donut chart	Risk _level; distinct count of id _student	Query 7
V4	Engagement vs. Performance	Clustered column	Engagement _level; average avg _score; students	Query 8
V5	Total Clicks vs. Average Score	Column or line	click band; average avg _score; students	Query 9
V6	Dropout Rate by Engagement Level	Bar chart	Engagement _level; dropout students; dropout rate (%)	Query 10
V7	Performance vs. Dropout	Bar chart	Performance _level; dropout students; dropout rate (%)	Query 11
V8	Regional Analysis	Map or bar, with table	region; students; average avg _score; average total _clicks	Query 12
V9	Educational Background Analysis	Table or bar	Highest _education; students; average avg _score; average total _clicks	Query 13

5. Dashboard

5.1 Dashboard Design



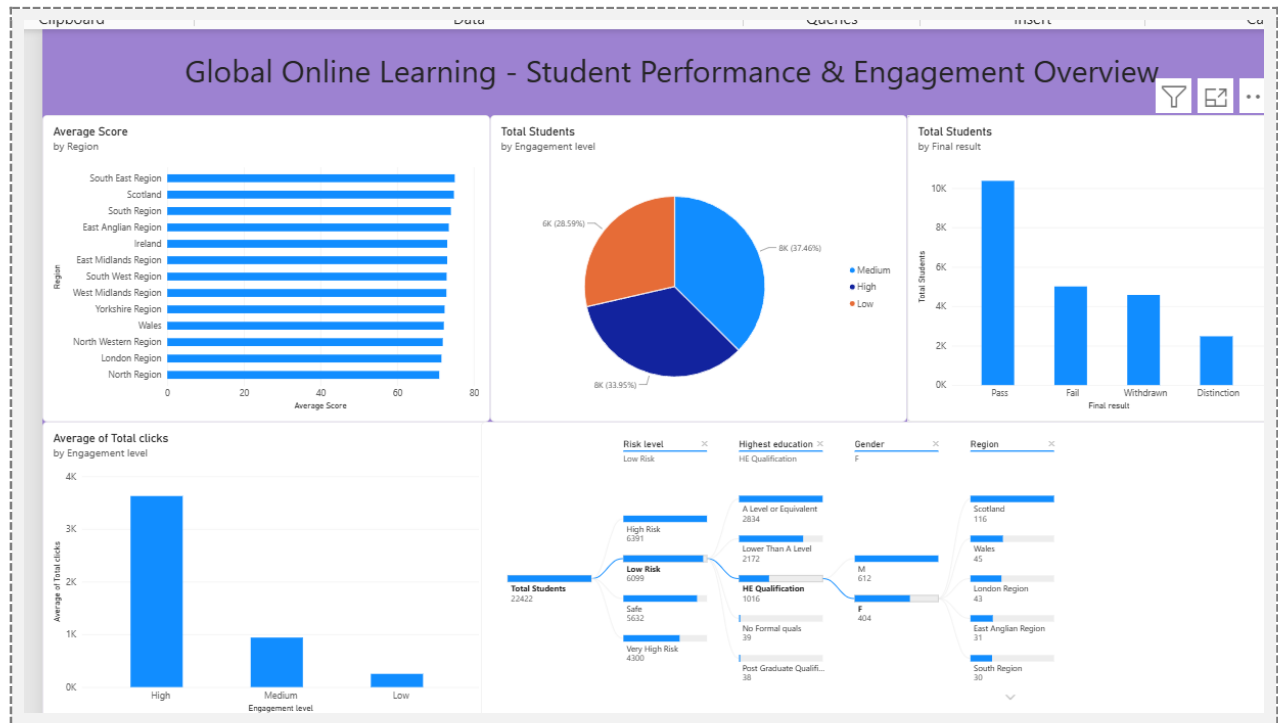
Layout: the top row holds five KPI cards (Total Students, Average Student Score, Average Total Clicks, Students Passed, Students Dropped Out), followed by a row of five slicers. Below are the engagement, performance and risk visuals, then the engagement-vs-performance and clicks-vs-score charts, the two dropout charts, and finally the regional and educational background views.

KPI Cards

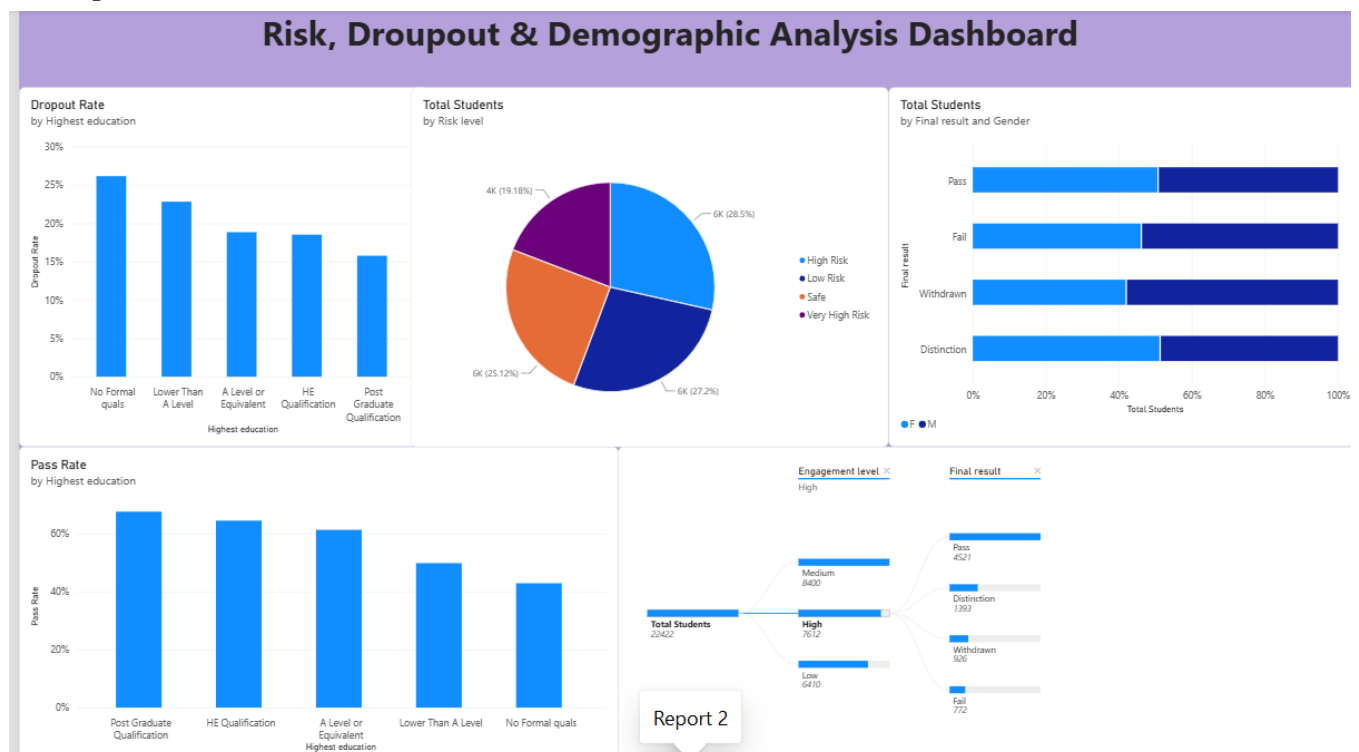


6. Report

6.1 Report 1



6.2 Report 2



7. Performance Testing

7.1 Utilization of Data Filters

No.	Filter (slicer)	Field	Applies to
1	Region	region	All visuals
2	Educational background	highest_education	All visuals
3	Engagement level	engagement_level	All visuals
4	Performance level	performance_level	All visuals
5	Risk level	risk_level	All visuals

Total data filters: 5 slicers.

7.2 No of Calculation Columns/Measures

No.	Name	Type	Logic
1	Total Students	Measure	Distinct count of id_student
2	Average Student Score	Measure	Average of avg_score, rounded to 2 decimals
3	Average Total Clicks	Measure	Average of total_clicks, rounded to 2 decimals
4	Students Passed	Measure	Count of records where pass_flag = 1
5	Students Dropped Out	Measure	Count of records where dropout_flag = 1
6	Dropout Rate (%)	Measure	Dropped-out students divided by total students, times 100
7	Click Band	Calculated column	Groups total_clicks into 0-500, 500-1000, 1000-2000, 2000-3000 and 3000+

7.3 No of Visualization

Type	Count	Details
KPI cards	5	Students, average score, average clicks, passed, dropped out
Charts and tables	9	V1 to V9 in Section 4.2
Slicers	5	Section 7.1
Total visuals on the dashboard	19	

8. Conclusion/Observation

The Online Learning Engagement Analytics solution brings student engagement, performance, risk and dropout into one dashboard and report. It answers the questions raised by the customer problem statements: instructors

can see which engagement levels lose students, and program managers can compare dropout and results across regions and educational backgrounds.

Key observations from the analysis:

Visual ID	Chart Component Name	Placement Location & Logic	Dashboard Live Value / Summary Metrics
V1	Engagement Level Distribution	Top-Right Donut Chart (Total Students by Platform Activity Level Mapping)	High Engagement Group: Highest Interaction Tracks Medium Engagement: Mid-level Clicks Profile Low Engagement Group: Critical Support Pool
V2	Performance Level Analysis	Center-Left Horizontal Bar Graph (Tracking Total Students by Regional Demographics)	Scotland: Top Enrolment Hub (~2.5K Students) East Anglian / South Region: Fast Growing Segments North Region: Smallest Student Concentration
V3	Risk Level Distribution	Center-Pie Chart Breakdown (Isolating Risk Profiles Across the 22K Student Ecosystem)	High Risk: 28.5% (6,391 Students) Low Risk: 27.2% (6,099 Students) Safe Status: 25.12% (5,632 Students) Very High Risk: 19.18% (4,300 Students)
V4	Engagement vs. Performance	Bottom-Left Column Chart (Average Score Distribution Indexed by Engagement Level)	High Engagement Grade: ~78 Average Score Medium Engagement Grade: ~74 Average Score Low Engagement Grade: ~70 Base Score
V5	Total Clicks vs. Average Score	Center Column Chart (Average Score Tracked by	High Performers: ~78 Average Grade Scale

		Performance Level Tier Mapping)	Medium Performers: ~60 Average Grade Scale Low Performers: ~38 Critical Grade Scale
V6	Dropout Rate by Engagement	Bottom decomposition branch tree tracks student volumes dropping out.	Highest Loss Category: Low platform interaction paths.
V7	Performance vs. Dropout	Vertical stacked segment chart tracking gender final pass profiles.	Pass / Distinction Rates: Significantly higher among PG and HE qualification backgrounds.
V8	Regional Demographics Analysis	Left sidebar bar mapping geographic locations with student counts.	Identifies support target groups in specific regional areas instantly.
V9	Educational Background Analysis	Center right tree tracking educational background distributions.	Top Demographic: A-Level or Equivalent & Lower Than A-Level paths.

9. Future Scope

- Build a predictive model that flags students likely to drop out before it happens.
- Add course-level and time-based views to see when engagement starts to fall.
- Refresh the dashboard automatically as new activity data arrives.
- Send alerts to instructors when a student moves into a high-risk level.
- Add row-level security so each instructor sees only their own students.

10. Appendix

10.1 Source Code (if any)

SQL queries used for the analysis (table: online _education _dataset).

1. Total Number of Students

```
SELECT
    COUNT (DISTINCT id _student) AS total _students
FROM online _education _dataset;
```

2. Average Student Score

```
SELECT
    ROUND (AVG (avg _score), 2) AS average _student _score
FROM online _education _dataset;
```

3. Average Total Clicks

```
SELECT
    ROUND (AVG (total _clicks), 2) AS average _total _clicks
FROM online _education _dataset;
```

4. Pass and Dropout Analysis

```
SELECT
    SUM (CASE WHEN pass _flag = 1 THEN 1 ELSE 0 END) AS students _passed,
    SUM (CASE WHEN dropout _flag = 1 THEN 1 ELSE 0 END) AS students _dropped _out
FROM online _education _dataset;
```

5. Engagement Level Analysis

```
SELECT
    Engagement _level,
    COUNT (DISTINCT id _student) AS total _students
FROM online _education _dataset
GROUP BY engagement _level
ORDER BY total _students DESC;
```

6. Performance Level Analysis

```
SELECT
    Performance _level,
    COUNT (DISTINCT id _student) AS total _students,
    ROUND (AVG (avg _score), 2) AS average _score
FROM online _education _dataset
GROUP BY performance _level
ORDER BY average _score DESC;
```

7. Risk Level Analysis

```
SELECT
    Risk _level,
    COUNT (DISTINCT id _student) AS total _students
FROM online _education _dataset
GROUP BY risk _level
ORDER BY total _students DESC;
```

8. Engagement vs. Performance

```
SELECT
    Engagement _level,
    ROUND (AVG (avg _score), 2) AS average _score,
    COUNT (DISTINCT id _student) AS total _students
FROM online _education _dataset
GROUP BY engagement _level
ORDER BY average _score DESC;
```

9. Total Clicks vs. Average Score

```
SELECT
    CASE WHEN total _clicks < 500 THEN '0 - 500' WHEN total _clicks < 1000 THEN '500 - 1000' WHEN
total _clicks < 2000 THEN '1000 - 2000' WHEN total _clicks < 3000 THEN '2000 - 3000' ELSE '3000+'
END AS click _band,
    COUNT (DISTINCT id _student) AS total _students,
    ROUND (AVG (avg _score), 2) AS average _score
FROM online _education _dataset
GROUP BY click _band
ORDER BY average _score DESC;
```

10. Dropout Analysis by Engagement Level

```
SELECT
    Engagement _level,
    COUNT (DISTINCT id _student) AS total _students,
    SUM (CASE WHEN dropout _flag = 1 THEN 1 ELSE 0 END) AS dropout _students,
    ROUND (100.0 * SUM (CASE WHEN dropout _flag = 1 THEN 1 ELSE 0 END) / COUNT (*), 2) AS dropout
_rate
FROM online _education _dataset
GROUP BY engagement _level
ORDER BY dropout _rate DESC;
```

11. Performance vs. Dropout

```
SELECT
    Performance _level,
    COUNT (DISTINCT id _student) AS total _students,
    SUM (CASE WHEN dropout _flag = 1 THEN 1 ELSE 0 END) AS dropout _students,
    ROUND (100.0 * SUM (CASE WHEN dropout _flag = 1 THEN 1 ELSE 0 END) / COUNT (*), 2) AS dropout
_rate
FROM online _education _ dataset
GROUP BY performance _ level
ORDER BY dropout _ rate DESC;
```

12. Regional Analysis

```
SELECT
    region,
    COUNT (DISTINCT id _student) AS total _students,
    ROUND (AVG (avg _score), 2) AS average _score,
    ROUND (AVG (total _ clicks), 2) AS average _clicks
FROM online _education _dataset
GROUP BY region
ORDER BY total _students DESC;
```

13. Educational Background Analysis

```
SELECT
    Highest _education,
    COUNT (DISTINCT id _student) AS total _students,
    ROUND (AVG (avg _score), 2) AS average _score,
    ROUND (AVG (total _clicks), 2) AS average _clicks
FROM online _education _dataset
GROUP BY highest _education
ORDER BY average _score DESC;
```